

Agenda:

- Design a parking lot
 - Intro to the problem
 - Gathering requirements
 - Class diagram
 - schema design

Design a parking lot

step 0 : Overview → known system
unknown system

1) Entity or an application

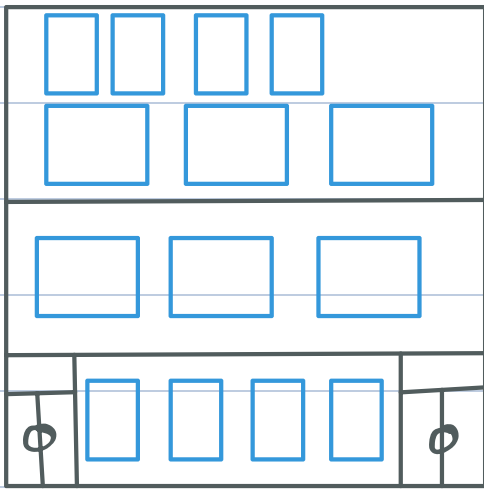
↳ Parking lot management system.

2) Persist memory → Yes → In-memory DB.

3) Input → Hard coded.

step 1 : Requirements Gathering

↳ visualise → Physical Structure
→ user journey.



- Each parking lot will have multiple floors.

- Each parking floor will have multiple slots.

- Each slot can accommodate different vehicle types.

- Each parking lot will have multiple entry & exit gates.

- A token will be generated at entry gate.
- Token should have vehicle & entry time.
optionally → slot no, gate, operator

Strategy

- Parking slot assignment will be done.
- Fee calculation & bill generation should be done at exit gate.

- Fee calculation can be dynamic.

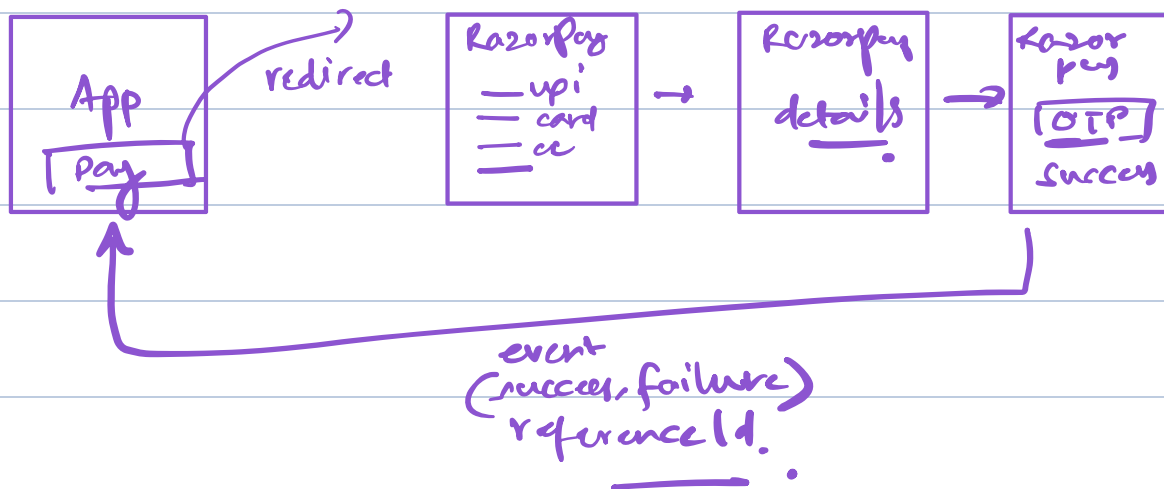
↳ Strategy

- Payment can be done in multiple modes like CASH, CARP, UPI, Fastag

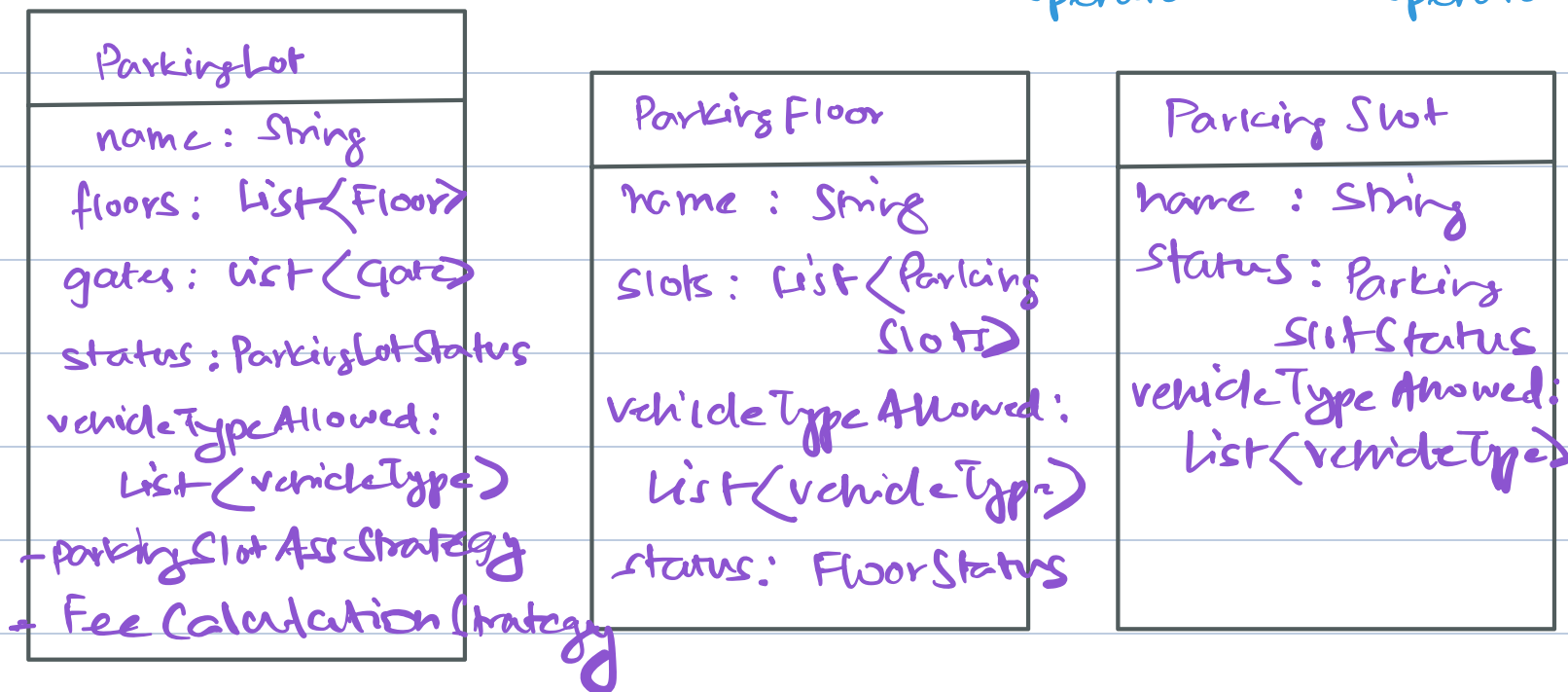
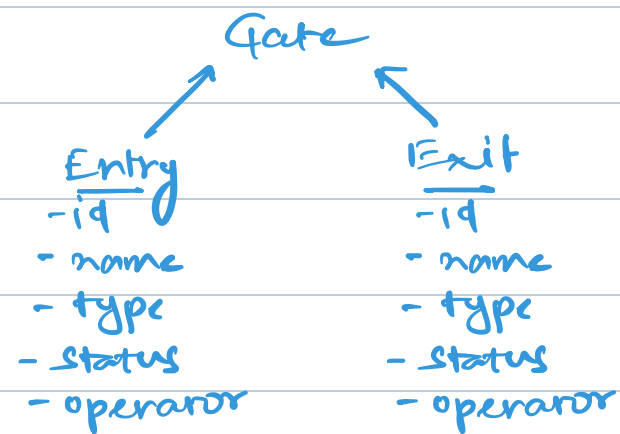
- Can there be multiple payment for the same bill.

- Can we allow refund of payment.

- Will be using 3rd party payment gateways.



Class Diagram



Gate	Operator	Vehicle
name: String	name: String	registration - no: String
operator: Operator	Gate: Gate	owner_name:
	emp_id: int	

Status: GateStatus
type: GateType

String

String
ownerNum:
Integer
VehicleType

Ticket
number: String
entryTime: DateTime
gate: Gate
operator: Operator
vehicle: Vehicle
floor: ParkingFloor
slot: parkingSlot

Bill
number: String
ticket: Ticket
exitTime: DateTime
amount: Int
gate: Gate
operator: operator

Payment
mode: PaymentMode
status: PaymentStatus
amount: Integer
transactionId: String

Status: BillStatus
List<Payment>

Schema Design

Parking Lot

enum

id | name | status | slotAssStrat | FeeCalcStrat

Parking Floor

id | name | status | parkingLotID

Parking Slot

id | name | status | parkingFloorID

PL : PF
1 : 1
1 : M

PF : PS
1 : 1
1 : M

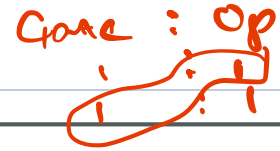
PL : Gate

Gate

id	name	status	GateType	ParkingLotID	operatorID
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Operator

id	name	employeeID
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vehicle

id	reg-no	owner-name	ownerContact	type
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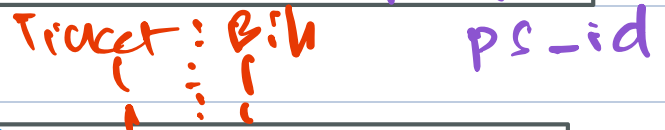
Ticket

id	number	entryTime	status	gateID	opID	pf-id	ps-id
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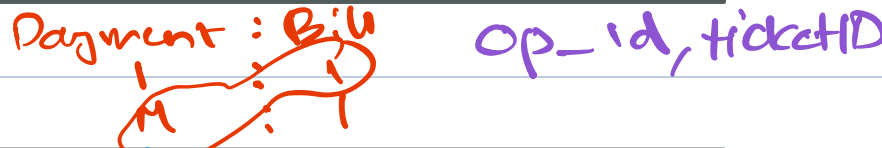
Bill

id	number	status	amount	exitTime	gateID
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Payment

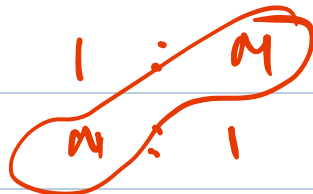
id	status	amount	mode	txn-id	Bill-id
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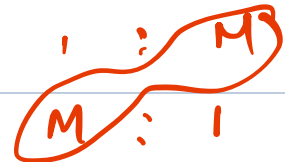
PL : VT



PF : VT



PS : VT



ParkingLot VehicleType Map

id	parkingLotID	vehicleType	count
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ParkingFloor Vehicle Type Map

id	parkingFloorId	vehicleType	count
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ParkingSlot Vehicle Type Map

id	parkingSlotId	vehicleType	count
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list < vehicle Type >

Map < vehicleType , Integer >

id	floor	vehType	count
1	1A	2W	10
2	2A	4W	20

Home work → code entity
classes.